

Exploring Cyclic Structures in Polynomial Rings with Applications to the Collatz Conjecture

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Abstract

We develop an algebraic framework for studying cyclic structures in polynomial rings with applications to the Collatz conjecture. We establish polynomial representations of natural numbers and define successor operations that preserve polynomial structure under Collatz-like transformations. Our main theoretical contributions include the Path Identity and Cycle Element Identity, which constrain the polynomial forms that can participate in cycles. The work culminates in a cyclotomic polynomial factorization of the key expression $d_p(g, h) = h^{e_p} - g^{o_p}$, revealing connections to algebraic number theory and providing tools for analyzing cycle structure in dynamical systems.

1 Introduction

1.1 The Collatz Conjecture

The Collatz conjecture, also known as the $3n + 1$ problem, is one of the most famous unsolved problems in mathematics. The conjecture states that for any positive integer n , the sequence defined by:

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases} \quad (1)$$

will eventually reach the cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$. Despite extensive computational verification and numerous partial results, a complete proof has remained elusive for over 80 years.

1.2 Motivation for Algebraic Approaches

Traditional approaches to the Collatz conjecture have largely focused on analytic number theory, probabilistic arguments, and computational methods. However, the discrete nature of the problem suggests that algebraic techniques, particularly those involving polynomial representations, might provide insights. By transforming the problem from the domain of integer sequences to polynomial algebra, we can leverage tools from algebraic geometry and commutative algebra.

The fundamental Path Identity $2^e x_n - 3^o x_0 = k$ and related Cycle Element Identity have been established in prior work [?]. What we develop here is a systematic polynomial framework that transforms these discrete relationships into algebraic constraints, enabling rigorous analysis of cycle structure.

1.3 Our Contributions

This paper explores cyclic structures in polynomial rings through:

- Polynomial representations $\sigma_p(u, v)$ establishing a bijection between natural numbers and the algebraic forms of cycle and path identities
- Successor operations that preserve polynomial structure under Collatz transformations
- Systematic analysis of the Path Identity and Cycle Element Identity (building on prior work)
- A cyclotomic factorization of $d_p(g, h) = h^{e_p} - g^{o_p}$

- The remarkable property that evaluating σ_p or k_p polynomials with appropriate arguments recovers the original integers that encode the identities of the algebraic forms

The novel contribution lies not in the identities themselves, but in the polynomial framework that transforms discrete dynamics into algebraic constraints, enabling systematic solution theory and revealing deep structural connections.

1.4 Paper Organization

Section 2 develops polynomial representations and establishes fundamental properties. Section 3 introduces successor operations and proves the Path Identity. Section 4 establishes the Cycle Element Identity and explores its implications. Section 5 presents the cyclotomic factorization and discusses its significance for cycle analysis.

2 Polynomial Representations

2.1 The natural number, p

Every natural number, p , can be represented as follows:

$$p = 2^{n_p} + \sum_{i \in \mathbb{Z}_{n_p}} b_{p,i} 2^i \quad (2)$$

where $(b_{p,i})_{i \in \mathbb{Z}_{n_p}}$ is the sequence of n_p bits forming the binary representation of $p - 2^{n_p}$.

It may seem somewhat arbitrary to treat the most significant bit, 2^{n_p} , separately from the bits of lesser significance, however, in the treatment that follows it turns out to be more convenient to do so. The main function of the most significant bit is to denote the precise length of the lower-order bit sequence and so disambiguate two bit sequences that differ only in the number of high-order even bits. For example: $p = 9 = 1001_2$ and $q = 17 = 10001_2$ uniquely represent bit sequences of length 3 and 4 respectively whereas without the presence of the top bit, the values integer encoding of 001_2 and 0001_2 would yield an ambiguous 1 in each case. So, by using the most significant bit to explicitly encode the length of the bit sequence, it is possible to precisely encode a bit sequence of exactly n_p bits as an integer and thus to distinguish each such integer from the integers representing any other bit sequence of different length or composition.

2.2 The series, $o_{p,i}$

Define the series, $o_{p,j}$, as:

$$o_{p,i} = \begin{cases} o_{p,i-1} + b_{p,i} & \text{if } i \in [0, n_p - 1], \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

$$o_p = o_{p,n_p-1} \quad (4)$$

It follows that:

$$o_{p,j} = \sum_{i \in \mathbb{Z}_{n_p}}^j b_{p,i} \quad (5)$$

$$k > i \implies o_{p,k} \geq o_{p,i} \quad (6)$$

$o_{p,i}$ counts the number of odd bits of p at bit position i or less.

$o_p + 1$ is the Hamming weight of p .

2.3 The series, $e_{p,i}$

Define the series, $e_{p,i}$, as:

$$e_{p,i} = \begin{cases} 1 - b_{p,0} & \text{if } i = 0, \\ e_{p,i-1} + 1 - b_{p,i} & \text{if } i \in [1, n_p - 1]. \end{cases} \quad (7)$$

$$e_p = e_{p, n_p - 1} \quad (8)$$

It follows that:

$$e_{p,j} = \sum_{i \in \mathbb{Z}_{n_p}}^j (1 - b_{p,i}) = j + 1 - o_{p,j} \quad (9)$$

$$k > i \implies e_{p,k} \geq e_{p,i} \quad (10)$$

$$i + 1 = e_{p,i} + o_{p,i} \quad (11)$$

$$n_p = o_{p, n_p - 1} + e_{p, n_p - 1} = o_p + e_p \quad (12)$$

$e_{p,i}$ counts the number of even bits of p at position i or less.

2.4 The sequence, $\sigma_{p,j}$

Define $\sigma_{p,j}$, as:

$$j \in \mathbb{Z}_{o_p} \quad (13)$$

$$\sigma_{p,j} \in \mathbb{Z}_{n_p} \quad (14)$$

$$\sigma_{p,j} = i \iff b_{p,i} = 1 \quad (15)$$

$$j < k \iff \sigma_{p,j} < \sigma_{p,k} \quad (16)$$

$\sigma_{p,j}$ labels the position, i , the j^{th} least-significant odd bit of $\{b_{p,i}\}$.
Observe that:

$$o_{p, \sigma_{p,j}} = j + 1 \quad (17)$$

$$\sigma_{p,0} = 0 \iff b_{p,0} = 1 \quad (18)$$

$$e_{p, \sigma_{p,j}} = \sigma_{p,j} + 1 - o_{p, \sigma_{p,j}} = \sigma_{p,j} - j \quad (19)$$

$$n_{p, \sigma_{p,j}} = o_{p, \sigma_{p,j}} + e_{p, \sigma_{p,j}} = \sigma_{p,j} + 1 \quad (20)$$

Define $\{\sigma_p\}$, as:

$$\{\sigma_p\} = \{\sigma_{p,j} | j \in \mathbb{Z}_{o_p}\} \quad (21)$$

2.5 Equivalent representations of p

The following definitions of p are equivalent:

$$p = 2^{n_p} + \sum_{i \in \mathbb{Z}_{n_p}} b_{p,i} \cdot 2^i \quad (22)$$

$$= 2^{n_p} + \sum_{i \in \{\sigma_p\}} 2^i \quad (23)$$

$$= 2^{n_p} + \sum_{j \in \mathbb{Z}_{o_p}} 2^{\sigma_{p,j}} \quad (24)$$

3 The successor operation, succ_p

Define $p_k = \text{succ}_p^{(k)}(p)$ as:

$$p_k = \text{succ}_p^{(k)}(p) \quad (25)$$

$$= 2^{n_p} + \sum_{i \in \mathbb{Z}_{n_p}} b_{p,i} \cdot 2^{i-k} \quad (26)$$

$$= 2^{n_p} + \sum_{i \in \{\sigma_p\}} 2^{i-k} \quad (27)$$

$$= 2^{n_p} + \sum_{j \in \mathbb{Z}_{o_p}} 2^{\sigma_{p,j}-k} \quad (28)$$

with:

$$i - k, \sigma_{p,j} - k \in \mathbb{Z}_{n_p} \quad (29)$$

In other words, the top $n_p - k$ bits of $\{b_{p,i}\}$ are shifted down by k -bits and the bottom k -bits of p are shifted into the top-most k bits or, alternatively, the bottom n_p bits of the binary representation of p are rotated downwards by k -bits, always leaving the top 2^{n_p} bit in place.

It can be shown that:

$$b_{p_k,i} = b_{p,i+k} \quad (30)$$

$$o_{p_k} = o_p \quad (31)$$

$$e_{p_k} = e_p \quad (32)$$

$$n_{p_k} = n_p \quad (33)$$

$$o_{p_k,i} = \begin{cases} o_{p,i+k} - o_{p,k-1} & \text{if } i < n_p - k, \\ o_{p,i+k-n_p} + (o_{p,n_p-1} - o_{p,k-1}) & \text{if } i \geq n_p - k. \end{cases} \quad (34)$$

$$\sigma_{p_k,j} = \sigma_{p,j+o_{p,k-1} \bmod o_p} - k \bmod n_p \quad (35)$$

4 The cycle, $\text{cycle}(p)$

Define $\text{cycle}(p)$ as:

$$\text{cycle}(p) = (\text{succ}_p^{(i)}(p))_{i \in \mathbb{Z}_{n_p}} \quad (36)$$

5 The polynomial, $\sigma_p(u, v)$

The polynomial $\sigma_p(u, v)$ of modular degree is defined as:

$$\sigma_p(u, v) = \sum_{i \in \mathbb{Z}_{n_p}} b_{p,i} \cdot u^{o_p - o_{p,i}} \cdot v^i \quad (37)$$

$$= \sum_{j \in \mathbb{Z}_{o_p}} u^{o_p - 1 - j} \cdot v^{\sigma_{p,j}} \quad (38)$$

$$= u^{o_p - 1} \cdot \sum_{j \in \mathbb{Z}_{o_p}} u^{-j} \cdot v^{\sigma_{p,j}} \quad (39)$$

with:

$$\deg_u(\sigma_p) = o_p - 1 \quad (40)$$

$$\deg_v(\sigma_p) = n_p - 1 \quad (41)$$

Observe that:

$$\sigma_p(1, 2) = 1^{o_p-1} \cdot \sum_{j \in \mathbb{Z}_{o_p}} 1^{-j} \cdot 2^{\sigma_p, j} = \sum_{j \in \mathbb{Z}_{o_p}} 2^{\sigma_p, j} \quad (42)$$

So:

$$p = 2^{n_p} + \sigma_p(1, 2) \quad (43)$$

Since p completely and uniquely determines $(n_p, \sigma_p(u, v))$ and $(n_p, \sigma_p(u, v))$ completely and uniquely determines p , a bijection exists between the set of natural numbers, $\{p\}$, and the set of tuples of the form $\{(n_p, \sigma_p(u, v))\}$.

6 The successor operation, succ_σ

Define $\text{succ}_\sigma(\sigma_p(u, v))$ as:

$$\text{succ}_\sigma(\sigma_p(u, v)) = u^{b_{p,0}} \cdot \sigma_p(u, v) \cdot v^{-1} \quad (44)$$

Lemma 1. $\sigma_{p_1}(u, v) = \sigma_{\text{succ}_p(p)}(u, v) = \text{succ}_\sigma(\sigma_p(u, v))$

Proof. Noting that $p_1 = \text{succ}_p(p)$, and $o_{p_1} = o_p$ then:

$$\sigma_{p_1}(u, v) = u^{o_p-1} \cdot \sum_{j \in \mathbb{Z}_{o_p}} u^{-j} \cdot v^{\sigma_{p_1, j}} \quad (45)$$

Substituting $\sigma_{p_1, j} = \sigma_{p, j+o_{p,0}} - 1$:

$$\sigma_{p_1}(u, v) = u^{o_p-1} \cdot \sum_{j \in \mathbb{Z}_{o_p}} u^{-j} \cdot v^{\sigma_{p, j+o_{p,0}}-1} \quad (46)$$

Reindexing with $\hat{j} = j + o_{p,0}$:

$$\sigma_{p_1}(u, v) = u^{o_p-1} \cdot \sum_{\hat{j} \in \mathbb{Z}_{o_p}} u^{-\hat{j}+o_{p,0}} \cdot v^{\sigma_{p, \hat{j}}-1} \quad (47)$$

Extracting factors of $u^{o_{p,0}}$ and v^{-1} :

$$\sigma_{p_1}(u, v) = u^{o_{p,0}} \cdot \left(u^{o_p-1} \cdot \sum_{\hat{j} \in \mathbb{Z}_{o_p}} u^{-\hat{j}} \cdot v^{\sigma_{p, \hat{j}}} \right) \cdot v^{-1} \quad (48)$$

Substituting $\sigma_p(u, v) = u^{o_p-1} \cdot \sum_{\hat{j} \in \mathbb{Z}_{o_p}} u^{-\hat{j}} \cdot v^{\sigma_{p, \hat{j}}}$, $b_{p,0} = o_{p,0}$:

$$\sigma_{p_1}(u, v) = u^{b_{p,0}} \cdot \sigma_p(u, v) \cdot v^{-1} \quad (49)$$

And noting that:

$$\text{succ}_\sigma(\sigma_p(u, v)) = u^{b_{p,0}} \cdot \sigma_p(u, v) \cdot v^{-1} \quad (50)$$

We conclude that:

$$\sigma_{p_1}(u, v) = \sigma_{\text{succ}_p(p)}(u, v) = \text{succ}_\sigma(\sigma_p(u, v)) \quad (51)$$

□

Lemma 2. $\sigma_{p_k}(u, v) = \text{succ}_\sigma^{(k)}(\sigma_p(u, v)) = u^{o_{p,k}-1} \cdot \sigma_p(u, v) \cdot v^{-k}$

Proof. The truth of this lemma can be established with an inductive argument that considers the base case and the inductive case.

Base case: $\sigma_{p_0}(u, v) = \text{succ}_\sigma^{(0)}(\sigma_p(u, v)) = u^{o_{p,-1}} \cdot \sigma_p(u, v) \cdot v^0$

Noting $o_{p,-1} = 0$:

$$u^{o_{p,-1}} \cdot \sigma_p(u, v) \cdot v^0 = 1 \cdot \sigma_p(u, v) \cdot 1 = \sigma_p(u, v) \quad (52)$$

Since $p_0 = p$, we see that the base case is true.

The inductive case: $\sigma_{p_{k+1}}(u, v) = \text{succ}_\sigma^{(k+1)}(\sigma_p(u, v)) = u^{o_{p,k}} \cdot \sigma_p(u, v) \cdot v^{-(k+1)}$

Assuming the hypothesis is true for k , then:

$$\sigma_{p_k}(u, v) = \text{succ}_\sigma^{(k)}(\sigma_p(u, v)) = u^{o_{p,k-1}} \cdot \sigma_p(u, v) \cdot v^{-k} \quad (53)$$

We need to prove:

$$\sigma_{p_{k+1}}(u, v) = \text{succ}_\sigma^{(k+1)}(\sigma_p(u, v)) = u^{o_{p,k}} \cdot \sigma_p(u, v) \cdot v^{-(k+1)} \quad (54)$$

According to the definition of succ_σ :

$$\text{succ}_\sigma(\sigma_{p_k}(u, v)) = u^{b_{p_k,0}} \cdot \sigma_{p_k}(u, v) \cdot v^{-1} \quad (55)$$

Assuming the hypothesis is true for p_k , and substituting for $\sigma_{p_k}(u, v)$:

$$\text{succ}_\sigma(\sigma_{p_k}(u, v)) = u^{b_{p_k,0}} \cdot u^{o_{p,k-1}} \cdot \sigma_p(u, v) \cdot v^{-k} \cdot v^{-1} \quad (56)$$

Noting that $b_{p_k,0} = b_{p,k}$:

$$\text{succ}_\sigma(\sigma_{p_k}(u, v)) = u^{b_{p,k}} \cdot u^{o_{p,k-1}} \cdot \sigma_p(u, v) \cdot v^{-k} \cdot v^{-1} \quad (57)$$

Noting that $o_{p,k} = o_{p,k-1} + b_{p,k}$:

$$\sigma_{p_{k+1}}(u, v) = \text{succ}_\sigma(\sigma_{p_k}(u, v)) = u^{o_{p,k}} \cdot \sigma_p(u, v) \cdot v^{-(k+1)} \quad (58)$$

Finally, noting that $\sigma_{p_k}(u, v) = \text{succ}_\sigma^{(k)}(\sigma_p(u, v))$:

$$\text{succ}_\sigma(\sigma_{p_k}(u, v)) = \text{succ}_\sigma(\text{succ}_\sigma^{(k)}(\sigma_p(u, v))) = \text{succ}_\sigma^{(k+1)}(\sigma_p(u, v)) \quad (59)$$

Hence:

$$\sigma_{p_{k+1}}(u, v) = \text{succ}_\sigma^{(k+1)}(\sigma_p(u, v)) = u^{o_{p,k}} \cdot \sigma_p(u, v) \cdot v^{-(k+1)} \quad (60)$$

Thus, the inductive hypothesis is confirmed.

Given that both the base case and inductive case have been confirmed, the lemma is also confirmed, namely:

$$\sigma_{p_k}(u, v) = \text{succ}_\sigma^{(k)}(\sigma_p(u, v)) = u^{o_{p,k-1}} \cdot \sigma_p(u, v) \cdot v^{-k} \quad (61)$$

□

7 The polynomial, $k_p(g, h)$

Define $k_p(g, h)$ as:

$$k_p(g, h) = \frac{\sigma_p(g \cdot h, h)}{h^{o_p-1}} \quad (62)$$

Therefore:

$$k_p(g, h) = \frac{(g \cdot h)^{o_p-1} \cdot \sum_{j \in \mathbb{Z}_{o_p}} (g \cdot h)^{-j} \cdot h^{\sigma_{p,j}}}{h^{o_p-1}} \quad (63)$$

$$k_p(g, h) = g^{o_p-1} \cdot \sum_{j \in \mathbb{Z}_{o_p}} g^{-j} \cdot h^{\sigma_{p,j}-j} \quad (64)$$

$$k_p(g, h) = g^{o_p} \cdot \sum_{j \in \mathbb{Z}_{o_p}} g^{-o_p, \sigma_{p,j}} \cdot h^{e_{p, \sigma_{p,j}}} \quad (65)$$

$$k_p(g, h) = \sum_{i \in \mathbb{Z}_{n_p}} b_{p,i} \cdot g^{o_p - o_{p,i}} \cdot h^{e_{p,i}} \quad (66)$$

Observe that:

$$k_p\left(\frac{u}{v}, v\right) \cdot v^{o_p-1} = \frac{\sigma_p\left(\frac{u}{v} \cdot v, v\right)}{v^{o_p-1}} \cdot v^{o_p-1} = \sigma_p(u, v) \quad (67)$$

$$2^{n_p} + k_p\left(\frac{1}{2}, 2\right) \cdot 2^{o_p-1} = 2^{n_p} + \sigma_p(1, 2) = p \quad (68)$$

8 The successor operation, succ_k

Define $\text{succ}_k(k_p(g, h))$ as:

$$\text{succ}_k(k_p(g, h)) = \begin{cases} g \cdot k_p(g, h) + h^{e_p} - g^{o_p} & , \text{ if } b_{p,0} = 1 \\ \frac{k_p(g, h)}{h} & , \text{ if } b_{p,0} = 0 \end{cases} \quad (69)$$

Lemma 3. $\text{succ}_k(k_p(g, h)) = \frac{(g \cdot h)^{b_{p,0}} \cdot k_p(g, h) + h \cdot b_{p,0} \cdot (h^{e_p} - g^{o_p})}{h}$

Proof. If $b_{p,0} = 1$:

$$\frac{(g \cdot h)^1 \cdot k_p(g, h) + h \cdot 1 \cdot (h^{e_p} - g^{o_p})}{h} = g \cdot k_p(g, h) + h^{e_p} - g^{o_p} \quad (70)$$

If $b_{p,0} = 0$:

$$\frac{(g \cdot h)^0 \cdot k_p(g, h) + h \cdot 0 \cdot (h^{e_p} - g^{o_p})}{h} = \frac{k_p(g, h)}{h} \quad (71)$$

□

Lemma 4. $\text{succ}_k(k_p(g, h)) = k_{\text{succ}_p(p)}(g, h)$

Proof.

$$\text{succ}_k(k_p(g, h)) = \frac{(g \cdot h)^{b_{p,0}} \cdot k_p(g, h) + h \cdot b_{p,0} \cdot (h^{e_p} - g^{o_p})}{h} \quad (72)$$

Substituting $g = \frac{u}{v}$, $h = v$:

$$\text{succ}_k\left(k_p\left(\frac{u}{v}, v\right)\right) = \frac{\left(\frac{u}{v} \cdot v\right)^{b_{p,0}} \cdot k_p\left(\frac{u}{v}, v\right) + v \cdot b_{p,0} \cdot (v^{e_p} - \left(\frac{u}{v}\right)^{o_p})}{v} \quad (73)$$

Multiplying both sides by v^{o_p-1} :

$$\text{succ}_k\left(k_p\left(\frac{u}{v}, v\right)\right) \cdot v^{o_p-1} = \frac{u^{b_{p,0}} \cdot k_p\left(\frac{u}{v}, v\right) \cdot v^{o_p-1} + b_{p,0} \cdot (v^{e_p+o_p} - u^{o_p})}{v} \quad (74)$$

Substituting $\sigma_p(u, v) = k_p\left(\frac{u}{v}, v\right) \cdot v^{o_p-1}$:

$$\text{succ}_k\left(k_p\left(\frac{u}{v}, v\right)\right) \cdot v^{o_p-1} = \frac{u^{b_{p,0}} \cdot \sigma_p(u, v) + b_{p,0} \cdot (v^{e_p+o_p} - u^{o_p})}{v} \quad (75)$$

However, given that these polynomials are polynomials of modular degree:

$$v^{e_p+o_p} = v^{n_p} = v^0 = 1 \quad (76)$$

and:

$$u^{o_p} = u^0 = 1 \quad (77)$$

Therefore:

$$\text{succ}_k\left(k_p\left(\frac{u}{v}, v\right)\right) \cdot v^{o_p-1} = u^{b_{p,0}} \cdot \sigma_p(u, v) \cdot v^{-1} = \text{succ}_\sigma(\sigma_p(u, v)) = \sigma_{\text{succ}_p(p)}(u, v) \quad (78)$$

Divide both sides by v^{o_p-1} :

$$\text{succ}_k\left(k_p\left(\frac{u}{v}, v\right)\right) = \frac{\sigma_{\text{succ}_p(p)}(u, v)}{v^{o_p-1}} \quad (79)$$

Substituting, $u = g \cdot h, v = h$:

$$\text{succ}_k(k_p(g, h)) = \frac{\sigma_{\text{succ}_p(p)}(g \cdot h, h)}{h^{o_p-1}} \quad (80)$$

We know that $\frac{\sigma_{\text{succ}_p(p)}(g \cdot h, h)}{h^{o_p-1}}$ is the definition of $k_{\text{succ}_p(p)}(g, h)$, so:

$$\text{succ}_k(k_p(g, h)) = k_{\text{succ}_p(p)}(g, h) \quad (81)$$

□

9 The successor operation, $\text{succ}_{x,a}$

Define the sequence $x_i(g, h)$ is defined by the successor operation

$$x_{i+1}(g, h) = \text{succ}_{x,a}(x_i(g, h)) = \begin{cases} \frac{x_i(g, h)}{h} & , \text{ if } b_{p,i} = 0 \\ g \cdot x_i(g, h) + a(g, h) & , \text{ if } b_{p,i} = 1 \end{cases} \quad (82)$$

$a(g, h)$ is a fixed polynomial in g and h and $(x_i(g, h))$ is a sequence of polynomials in g and h whose successive elements satisfy the successor relationship defined by the bits of p , $(b_{p,i})_{i \in \mathbb{Z}_{n_p}}$.

It should be noted that both $a(g, h)$ or $(x_i(g, h))$ are unknowns which, if they exist, must satisfy the constraints of the successor operation.

Lemma 5. $x_{i+1}(g, h) = \frac{(g \cdot h)^{b_{p,i}} \cdot x_i(g, h) + a(g, h) \cdot b_{p,i} \cdot h}{h}$

Proof. **Case 1:** $b_{p,i} = 0$

$$x_{i+1}(g, h) = \frac{(g \cdot h)^0 \cdot x_i(g, h) + a(g, h) \cdot 0 \cdot h}{h} = \frac{x_i(g, h)}{h} \quad (83)$$

Case 2: $b_{p,i} = 1$

$$x_{i+1}(g, h) = \frac{(g \cdot h)^1 \cdot x_i(g, h) + a(g, h) \cdot 1 \cdot h}{h} = g \cdot x_i(g, h) + a(g, h) \quad (84)$$

Thus:

$$x_{i+1}(g, h) = \frac{(g \cdot h)^{b_{p,i}} \cdot x_i(g, h) + a(g, h) \cdot b_{p,i} \cdot h}{h} \quad (85)$$

□

10 The path identity

Theorem 1 (Path Identity). $x_k(g, h) \cdot h^{e_{p,k-1}} = g^{o_{p,k-1}} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^{k-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right)$

The path identity relates the value of the k^{th} element of the sequence $x_k(g, h)$ to the initial element $x_0(g, h)$ and the cumulative effects of each operation as determined by the bits $(b_{p,i})_{i \in [0, k-1]}$.

Proof. **Base case:** $x_1(g, h) \cdot h^{e_{p,0}} = g^{o_{p,0}} \cdot (x_0(g, h) + a(g, h) \cdot \sum_{i=0}^0 b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}})$

Subcase 1: $b_{p,0} = 0$

$$x_1(g, h) \cdot h^1 = g^0 \cdot (x_0(g, h) + a(g, h) \cdot 0 \cdot g^{-0} \cdot h^1) \quad (86)$$

$$x_1(g, h) = \frac{x_0(g, h)}{h} \quad (87)$$

Subcase 2: $b_{p,0} = 1$

$$x_1(g, h) \cdot h^0 = g^1 \cdot (x_0(g, h) + a(g, h) \cdot 1 \cdot g^{-1} \cdot h^0) \quad (88)$$

$$x_1(g, h) = g \cdot x_0(g, h) + a(g, h) \quad (89)$$

So, according to the definition for $x_{i+1}(g, h)$ the hypothesis holds for the base case.

Inductive case: $x_{k+1}(g, h) \cdot h^{e_{p,k}} = g^{o_{p,k}} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^k b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right)$

According to the definition of $x_{i+1}(g, h)$:

$$x_{k+1}(g, h) = \frac{(g \cdot h)^{b_{p,k}} \cdot x_k(g, h) + a(g, h) \cdot b_{p,k} \cdot h}{h} \quad (90)$$

Assuming the truth of the hypothesis, substitute for $x_k(g, h)$:

$$x_{k+1}(g, h) = \frac{(g \cdot h)^{b_{p,k}} \cdot \frac{g^{o_{p,k-1}} \cdot (x_0(g, h) + a(g, h) \cdot \sum_{i=0}^{k-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}})}{h^{e_{p,k-1}}} + a(g, h) \cdot b_{p,k} \cdot h}{h} \quad (91)$$

Redistributing powers of h :

$$x_{k+1}(g, h) = \frac{g^{b_{p,k}} \cdot g^{o_{p,k-1}} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^{k-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right) + a(g, h) \cdot b_{p,k} \cdot h^{e_{p,k-1}+1-b_{p,k}}}{h^{e_{p,k-1}+1-b_{p,k}}} \quad (92)$$

Combining powers of g and h :

$$x_{k+1}(g, h) = \frac{g^{o_{p,k}} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^{k-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right) + a(g, h) \cdot b_{p,k} \cdot h^{e_{p,k}}}{h^{e_{p,k}}} \quad (93)$$

Extend sum to include trailing term:

$$x_{k+1}(g, h) = \frac{g^{o_{p,k}} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^k b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right)}{h^{e_{p,k}}} \quad (94)$$

Multiply both sides by $h^{e_{p,k}}$:

$$x_{k+1}(g, h) \cdot h^{e_{p,k}} = g^{o_{p,k}} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^k b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right) \quad (95)$$

Hence:

$$x_k(g, h) \cdot h^{e_{p,k-1}} = g^{o_{p,k-1}} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^{k-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right) \quad (96)$$

□

11 The cycle element identity

Theorem 2 (Cycle Element Identity). $x_{q,a}(g, h) \cdot d_p(g, h) = a(g, h) \cdot k_q(g, h)$

Consider the case where $k = n_p$, the path identity becomes:

$$x_{n_p}(g, h) \cdot h^{e_{p,n_p-1}} = g^{o_{p,n_p-1}} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^{n_p-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right) \quad (97)$$

Simplifying the expressions for e_{p,n_p-1}, o_{p,n_p-1} :

$$x_{n_p}(g, h) \cdot h^{e_p} = g^{o_p} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^{n_p-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right) \quad (98)$$

If we require that $x_{n_p} = x_0$, then:

$$x_0(g, h) \cdot h^{e_p} = g^{o_p} \cdot \left(x_0(g, h) + a(g, h) \cdot \sum_{i=0}^{n_p-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right) \quad (99)$$

Subtract $g^{o_p} \cdot x_0$ from both sides:

$$x_0(g, h) \cdot h^{e_p} - g^{o_p} \cdot x_0(g, h) = g^{o_p} \cdot \left(a(g, h) \cdot \sum_{i=0}^{n_p-1} b_{p,i} \cdot g^{-o_{p,i}} \cdot h^{e_{p,i}} \right) \quad (100)$$

Re-arrange:

$$x_0(g, h) \cdot (h^{e_p} - g^{o_p}) = a(g, h) \cdot \sum_{i=0}^{n_p-1} b_{p,i} \cdot g^{o_p-o_{p,i}} \cdot h^{e_{p,i}} \quad (101)$$

Observe that:

$$\sum_{i=0}^{n_p-1} b_{p,i} \cdot g^{o_p - o_{p,i}} \cdot h^{e_{p,i}} = k_p(g, h) \quad (102)$$

Define:

$$d_p(g, h) = h^{e_p} - g^{o_p} \quad (103)$$

Finally, relabel $x_0(g, h)$ as $x_{p,a}(g, h)$, to yield:

$$x_{p,a}(g, h) \cdot d_p(g, h) = a(g, h) \cdot k_p(g, h) \quad (104)$$

More generally, for each i , there exists $q = \text{succ}_p^{(i)}(p)$ such that:

$$x_{q,a}(g, h) = x_i(g, h) = \text{succ}_{x,a}^{(i)}(x_0(g, h)) \quad (105)$$

By replacing p with q for any $q \in \text{cycle}(p)$:

$$x_{q,a}(g, h) \cdot d_q(g, h) = a(g, h) \cdot k_q(g, h) \quad (106)$$

but we know that $d_p(g, h) = d_q(g, h)$ for $q \in \text{cycle}(p)$, so:

$$x_{q,a}(g, h) \cdot d_p(g, h) = a(g, h) \cdot k_q(g, h) \quad (107)$$

Lemma 6. $\deg_g(a(g, h)) = \deg_g(x_{q,a}(g, h)) + 1$

Proof. Apply the $\deg_g$ function to both sides of the cycle element identity:

$$\deg_g(a(g, h) \cdot k_q(g, h)) = \deg_g(x_{q,a}(g, h) \cdot d_p(g, h)) \quad (108)$$

By the additivity under multiplication property of the $\deg_g$ function:

$$\deg_g(a(g, h)) + \deg_g(k_q(g, h)) = \deg_g(x_{q,a}(g, h)) + \deg_g(d_p(g, h)) \quad (109)$$

Substituting in the known values of $\deg_g(k_q(g, h))$ and $\deg_g(d_p(g, h))$:

$$\deg_g(a(g, h)) + o_p - 1 = \deg_g(x_{q,a}(g, h)) + o_p \quad (110)$$

Subtracting $o_p - 1$ from both sides:

$$\deg_g(a(g, h)) = \deg_g(x_{q,a}(g, h)) + 1 \quad (111)$$

□

12 Solutions

A solution is defined as a sequence of polynomials, $(x_{q,a}(g, h))_{q \in \text{cycle}(p)}$ for which the following are both true:

$$x_{\text{succ}_p(q),a}(g, h) = \text{succ}_{x,a}(x_{q,a}(g, h)), \quad \forall q \in \text{cycle}(p) \quad (112)$$

$$x_{q,a}(g, h) \cdot d_p(g, h) = a(g, h) \cdot k_q(g, h), \quad \forall q \in \text{cycle}(p) \quad (113)$$

That is, the successor relationship holds and the cycle element identity holds for each element q in $\text{cycle}(p)$.

Lemma 7. *At least one solution exists for each p .*

Proof. To see this, set:

$$a(g, h) = d_p(g, h) \quad (114)$$

$$x_{q,a}(g, h) = k_q(g, h) \quad (115)$$

which means that the cycle element identity is trivially satisfied.

Observe also that:

$$\text{succ}_{x,a} = \text{succ}_k \quad (116)$$

We know that:

$$k_{\text{succ}_p(q)}(g, h) = \text{succ}_k(k_q(g, h)) \quad (117)$$

We can substitute $\text{succ}_{x,a} = \text{succ}_k$, $x_{q,a}(g, h) = k_q(g, h)$, $x_{\text{succ}_p(q),a}(g, h) = k_{\text{succ}_p(q)}(g, h)$ to yield:

$$x_{\text{succ}_p(q),a}(g, h) = \text{succ}_{x,a}(x_{q,a}(g, h)) \quad (118)$$

□

12.1 The general solution

The general solution does not assume $a(g, h) = d_p(g, h)$.

Let's define:

$$\hat{a}(g, h) = \gcd(d_p(g, h), a(g, h)) \quad (119)$$

$$\bar{a}(g, h) = \frac{a(g, h)}{\hat{a}(g, h)} \quad (120)$$

$$f_{\hat{a}}(g, h) = \frac{d_p(g, h)}{\hat{a}(g, h)} \quad (121)$$

Then we can rewrite:

$$d_p(g, h) = f_{\hat{a}}(g, h) \cdot \hat{a}(g, h) \quad (122)$$

$$a(g, h) = \bar{a}(g, h) \cdot \hat{a}(g, h) \quad (123)$$

Lemma 8. $\gcd(f_{\hat{a}}(g, h), \bar{a}(g, h)) = 1$

Proof.

$$\hat{a}(g, h) = \gcd(d_p(g, h), a(g, h)) \quad (124)$$

$$= \gcd(f_{\hat{a}}(g, h) \cdot \hat{a}(g, h), \bar{a}(g, h) \cdot \hat{a}(g, h)) \quad (125)$$

$$= \hat{a}(g, h) \cdot \gcd(f_{\hat{a}}(g, h), \bar{a}(g, h)) \quad (126)$$

This implies $\gcd(f_{\hat{a}}(g, h), \bar{a}(g, h)) = 1$. □

Lemma 9. $\bar{a}(g, h) \mid x_{q,a}(g, h)$

Proof. Starting with the cycle element identity:

$$x_{q,a}(g, h) \cdot d_p(g, h) = a(g, h) \cdot k_q(g, h) \quad (127)$$

Substituting for $d_p(g, h)$ and $a(g, h)$ and dividing both sides by $\hat{a}(g, h)$:

$$x_{q,a}(g, h) \cdot f_{\hat{a}}(g, h) = \bar{a}(g, h) \cdot k_q(g, h) \quad (128)$$

Solving for $k_q(g, h)$:

$$k_q(g, h) = \frac{x_{q,a}(g, h) \cdot f_{\hat{a}}(g, h)}{\bar{a}(g, h)} \quad (129)$$

But we know that $k_q(g, h)$ is polynomial with integer coefficients and $\gcd(f_{\hat{a}}(g, h), \bar{a}(g, h)) = 1$, therefore $\bar{a}(g, h) \mid x_{q,a}(g, h)$. □

Lemma 10. $\exists \hat{x}_{q,a}(g, h) : x_{q,a}(g, h) = \bar{a}(g, h) \cdot \hat{x}_{q,a}(g, h)$

Proof. Given that $\bar{a}(g, h) \mid x_{q,a}(g, h)$, we can define the polynomial $\hat{x}_{q,\hat{a}}(g, h)$ as:

$$\hat{x}_{q,\hat{a}}(g, h) = \frac{x_{q,a}(g, h)}{\bar{a}(g, h)} \quad (130)$$

such that:

$$x_{q,a}(g, h) = \bar{a}(g, h) \cdot \hat{x}_{q,\hat{a}}(g, h) \quad (131)$$

□

Lemma 11. $f_{\hat{a}}(g, h) \mid k_q(g, h)$

Proof. Substituting for $x_{q,a}(g, h)$ and $\bar{a}(g, h)$:

$$k_q(g, h) = \hat{x}_{q,\hat{a}}(g, h) \cdot f_{\hat{a}}(g, h) \quad (132)$$

Thus:

$$f_{\hat{a}}(g, h) \mid k_q(g, h) \quad (133)$$

□

Lemma 12. $(\hat{x}_{q,\hat{a}}(g, h))_{q \in \text{cycle}(p)}$ is a solution.

Proof. First, substitute expressions for $a(g, h)$ and $x_{q,a}(g, h)$ into cycle element identity:

$$\bar{a}(g, h) \cdot \hat{x}_{q,a}(g, h) \cdot d_p(g, h) = \bar{a}(g, h) \cdot \hat{a}(g, h) \cdot k_q(g, h) \quad (134)$$

Dividing both sides by $\bar{a}(g, h)$:

$$\hat{x}_{q,a}(g, h) \cdot d_p(g, h) = \hat{a}(g, h) \cdot k_q(g, h) \quad (135)$$

The successor relationship is:

$$x_{i+1}(g, h) = \frac{(g \cdot h)^{b_{p,i}} \cdot x_i(g, h) + a(g, h) \cdot b_{p,i} \cdot h}{h} \quad (136)$$

which can be re-expressed as:

$$x_{\text{succ}_p^{(i+1)}(p),a}(g, h) = \frac{(g \cdot h)^{b_{p,i}} \cdot x_{\text{succ}_p^{(i)}(p),a}(g, h) + a(g, h) \cdot b_{p,i} \cdot h}{h} \quad (137)$$

and with $q = \text{succ}_p^{(i)}(p)$:

$$x_{\text{succ}_p(q),a}(g, h) = \frac{(g \cdot h)^{b_{q,0}} \cdot x_{q,a}(g, h) + a(g, h) \cdot b_{q,0} \cdot h}{h} \quad (138)$$

with the substitutions $a(g, h) = \bar{a}(g, h) \cdot \hat{a}(g, h)$ and $x_{q,a}(g, h) = \bar{a}(g, h) \cdot \hat{x}_{q,\hat{a}}(g, h)$:

$$\bar{a}(g, h) \cdot \hat{x}_{\text{succ}_p(q),\hat{a}}(g, h) = \frac{(g \cdot h)^{b_{q,0}} \bar{a}(g, h) \cdot \hat{x}_{q,\hat{a}}(g, h) + \bar{a}(g, h) \cdot \hat{a}(g, h) \cdot b_{q,0} \cdot h}{h} \quad (139)$$

Dividing both sides by $\bar{a}(g, h)$:

$$\hat{x}_{\text{succ}_p(q),\hat{a}}(g, h) = \frac{(g \cdot h)^{b_{q,0}} \hat{x}_{q,\hat{a}}(g, h) + \hat{a}(g, h) \cdot b_{q,0} \cdot h}{h} \quad (140)$$

Therefore, $(\hat{x}_{q,\hat{a}}(g, h))_{q \in \text{cycle}(p)}$ is also a solution. □

Lemma 13. $(x_{q,a}(g, h))_{q \in \text{cycle}(p)}$ is a solution $\implies \deg_g(\gcd(a(g, h), d_p(g, h))) \geq 1$

Proof. By a previous lemma $\deg_g(\hat{a}(g, h)) \geq 1$ which therefore implies that $\deg_g(\gcd(a(g, h), d_p(g, h))) \geq 1$.

Furthermore, for every solution $(x_{q,a}(g, h))_{q \in \text{cycle}(p)}$ where $a(g, h) \nmid d_p(g, h)$ there is a (not necessarily unique) solution $(\hat{x}_{q,\hat{a}}(g, h))_{q \in \text{cycle}(p)}$ where:

$$d_p(g, h) = f_{\hat{a}}(g, h) \cdot \hat{a}(g, h) \quad (141)$$

and:

$$k_q(g, h) = f_{\hat{a}}(g, h) \cdot \hat{x}_{q, \hat{a}}(g, h), \quad \forall q \in \text{cycle}(p) \quad (142)$$

□

Lemma 14. $f_{\hat{a}}(g, h) \mid d_p(g, h) \wedge f_{\hat{a}}(g, h) \mid k_q(g, h) \implies f_{\hat{a}}(g, h) \mid k_{\text{succ}_p(q)}(g, h), \forall q \in \text{cycle}(p)$

Proof.

$$k_{\text{succ}_p(q)}(g, h) = \text{succ}_k(k_q(g, h)) = \frac{(g \cdot h)^{b_{q,0}} \cdot k_q(g, h) + h \cdot b_{q,0} \cdot d_p(g, h)}{h} \quad (143)$$

$$\text{lc}_g(d_p(g, h)) = -1 \implies h \nmid d_p(g, h) \wedge h \nmid \hat{a}(g, h) \wedge h \nmid f_{\hat{a}}(g, h) \quad (144)$$

$$h \mid k_q(g, h) \implies h \mid f_{\hat{a}}(g, h) \cdot x_{q, \hat{a}}(g, h) \wedge h \nmid f_{\hat{a}}(g, h) \implies h \mid x_{q, \hat{a}}(g, h) \quad (145)$$

$$k_{\text{succ}_p(q)}(g, h) = f_{\hat{a}}(g, h) \cdot \left(\frac{(g \cdot h)^{b_{q,0}} \cdot x_{q, \hat{a}}(g, h) + h \cdot b_{q,0} \cdot \hat{a}(g, h)}{h} \right) \quad (146)$$

Since $h \nmid f_{\hat{a}}(g, h)$, we have $f_{\hat{a}}(g, h) \mid k_{\text{succ}_p(q)}(g, h)$. □

12.2 The minimal solution, $(\hat{x}_{q, \hat{a}_{\min}}(g, h))_{q \in \text{cycle}(p)}$

$\hat{a}_{\min}(g, h)$ is the divisor of $d_p(g, h)$ which maximizes $f_{\hat{a}_{\min}}(g, h)$ such that $f_{\hat{a}_{\min}}(g, h)$ also divides $k_p(g, h)$.

Therefore, we require:

$$f_{\hat{a}_{\min}}(g, h) = \gcd(d_p(g, h), k_q(g, h)), \quad \forall q \in \text{cycle}(p) \quad (147)$$

such that:

$$d_p(g, h) = f_{\hat{a}_{\min}}(g, h) \cdot \hat{a}_{\min}(g, h) \quad (148)$$

Lemma 15. $\gcd(k_p(g, h), d_p(g, h)) = \gcd(k_q(g, h), d_p(g, h)), \forall q \in \text{cycle}(p)$

Proof. By the previous lemma we know that:

$$\gcd(k_p(g, h), d_p(g, h)) \mid k_q(g, h) \wedge \gcd(k_q(g, h), d_p(g, h)) \mid k_p(g, h) \quad (149)$$

This is a system of equations of the form:

$$\gcd(a, b) \mid c \quad (150)$$

$$\gcd(a, c) \mid b \quad (151)$$

Suppose $\gcd(a, b) \neq \gcd(a, c)$, then:

$$\exists f : f \mid \gcd(a, b) \wedge f \nmid \gcd(a, c) \quad (152)$$

But:

$$f \mid \gcd(a, b) \implies f \mid c \implies f \mid \gcd(a, c) \quad (153)$$

which is a contradiction.

In other words:

$$f_{\hat{a}_{\min}}(g, h) = \gcd(k_p(g, h), d_p(g, h)) = \gcd(k_q(g, h), d_p(g, h)), \quad \forall q \in \text{cycle}(p) \quad (154)$$

□

13 The polynomial factors of $d_p(g, h)$

$$d_p(g, h) = h^{e_p} - g^{o_p} \quad (155)$$

$$= g^{o_p} \cdot \left(\frac{h^{e_p}}{g^{o_p}} - 1 \right) \quad (156)$$

$$= g^{o_p} \cdot (x^c - 1) \quad (157)$$

$$= g^{\frac{o_p}{c} \cdot c} \cdot (x^c - 1) \quad (158)$$

where

$$c = \gcd(e_p, o_p) \quad (159)$$

$$x = \left(\frac{h^{\frac{e_p}{c}}}{g^{\frac{o_p}{c}}} \right) \quad (160)$$

$$x^c - 1 = \prod_{q|c} \Phi_q(x) \quad (161)$$

$$c = \sum_{q|c} \varphi(q) \quad (162)$$

$$g^{\frac{o_p}{c} \cdot c} = g^{\frac{o_p}{c} \cdot \sum_{q|c} \varphi(q)} = \prod_{q|c} g^{\frac{o_p}{c} \cdot \varphi(q)} \quad (163)$$

where $\Phi_q(x)$ is the q^{th} cyclotomic polynomial and $\varphi(q)$ is the Euler totient function. Therefore:

$$d_p(g, h) = \prod_{q|c} g^{\frac{o_p}{c} \cdot \varphi(q)} \cdot \Phi_q\left(\frac{h^{\frac{e_p}{c}}}{g^{\frac{o_p}{c}}}\right), \quad c = \gcd(e_p, o_p) \quad (164)$$

14 Conclusion

14.1 Summary of Results

We have developed an algebraic framework for studying cyclic structures in polynomial rings with applications to the Collatz conjecture. The key contributions of this work are:

1. **Polynomial Representation:** Every natural number p admits a representation as a bivariate polynomial $\sigma_p(u, v)$ that encodes its binary structure, establishing a bijection between natural numbers and the algebraic forms of cycle and path identities.
2. **Successor Operations:** We have defined polynomial operations that preserve polynomial structure under Collatz-like transformations, enabling algebraic analysis of sequence dynamics.
3. **Polynomial Framework for Known Identities:** Building on prior work establishing the Path Identity and Cycle Element Identity, we transform these into polynomial constraints $x_{q,a}(g, h) \cdot d_p(g, h) = a(g, h) \cdot k_q(g, h)$.
4. **Evaluation Property:** The remarkable feature that evaluating σ_p or k_p polynomials with appropriate arguments and scaling factors recovers the original integers that encode the identities of the algebraic forms.
5. **Cyclotomic Factorization:** The expression $d_p(g, h) = h^{e_p} - g^{o_p}$ factors as a product of cyclotomic polynomials, revealing connections to algebraic number theory.

14.2 Implications for the Collatz Conjecture

The polynomial framework transforms the discrete dynamics of the Collatz sequence into polynomial algebra, where algebraic tools can be applied. The Cycle Element Identity, in particular, places restrictions on the polynomial forms that can satisfy cycle conditions. Combined with the cyclotomic factorization, this suggests that non-trivial cycles may be algebraically impossible.

The framework addresses the "finite cycles" aspect of the Collatz conjecture by providing algebraic constraints that could prove the uniqueness of the $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ cycle. However, the "no-escape" problem—proving that all trajectories eventually reach a cycle—remains an open challenge requiring additional techniques.

14.3 Future Work

Several promising directions emerge from this work:

- **Computational Implementation:** Developing algorithms to compute polynomial representations efficiently and verify the theoretical predictions.
- **Extension to Generalized Problems:** Applying the framework to $(gx + a, x/h)$ generalizations of the Collatz problem.
- **Alternative Polynomial Representations:** Exploring relationships with other polynomial representations of natural numbers, including gh-forms (to be introduced in future work). Note that current gh-forms results establish symbolic polynomial uniqueness but do not yet preclude numerical evaluation equality at specific values like $g = 3, h = 2$.
- **Algebraic Geometry Applications:** Investigating the geometric properties of the polynomial varieties defined by cycle conditions.
- **No-Escape Problem:** Developing complementary techniques to address trajectory convergence.
- **Strengthening GH-Forms Results:** Extending gh-forms uniqueness theorems from symbolic polynomial equality to numerical evaluation equality at concrete values, which is necessary for complete cycle uniqueness proofs.

The polynomial framework opens avenues for attacking one of mathematics' most enduring problems, providing algebraic perspectives on the deceptively simple $3n + 1$ dynamics.

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